Aalap 3.0

QA report

Build: Aalap 3.0.0 (versionCode 30), release build, file dist/Aalap-3.0.apk
Tested on: Samsung (Android, One UI), signed in to Spotify and YouTube Music, 23 Oct 2026
Tested by: Claude, acting as senior QA engineer: every screen driven on the phone, background jobs and saved data inspected, failure cases checked, bugs fixed and re-tested.

1. Summary

Areas tested	16 (Home, Aaj ka Aalap, Quick picks, charts, moods, mixes, Search, Library, playlists, artist and album pages, player, mini player, Liked Songs, History/Downloads/Local files, Profile and Settings, background sync and taste)
Bugs found and fixed	**24** (2 data bugs, 6 behaviour bugs, 4 failure-case bugs, 3 accessibility, 9 display/text)
Open bugs	**0** known
Things not run on the phone (and why)	5, listed in section 6
Player close smoothness	worst frame **~30 ms** (was 150–450 ms before this round of work)

Everything listed in section 3 as **PASS** was done on the phone and checked by reading the screen, the playback state or the app's saved data.

2. How Aalap works (feature by feature)

Home

- **Greeting** follows the time of day (Good morning / Late-night listening).
- **Aaj ka Aalap poster**: one of **50 handmade craft designs**, a new one at midnight (also while the app is open). The shuffle button picks another design for the rest of the day. Tapping the poster opens **today's 30 songs**: 8 trending, 7 top hits, 9 for your taste, 6 of your likes, rebuilt every day.
- **Quick picks**: trending songs mixed with picks for your taste; a new set every day **and** on every refresh (pull down, or double-tap the Home tab). Each row shows why it's there (Trending / Top hit / Your taste) and a gold heart if you liked it.
- **Mood stations**: 17 moods, each ordered for your taste; tap to play. They change every 6 hours.
- **Today's charts**: Top 50 India, Top 50 Global, Trending India (Spotify's *Trending Now India*), Trending Global (*Viral Hits*). Cached once a day.
- **Made for you**: Daily Mix 1–3 from your top artists, new every day.
- **Jump back in, Recently played, Your all-time favourites, Your artist radios, New from your artists, Popular albums and singles, More of what you like.**
- Home also rebuilds by itself when a new 6-hour window or a new day starts, even if left open.

Search

- 10 **trending searches** (5 from YouTube Music, 5 from Spotify), refreshed daily; mood and explore tiles.
- Results: songs, artists, albums, each shown once, with **All / Songs / Artists / Albums** filters and taste tags.

Library

- Tiles: **Liked Songs** (same count everywhere), Downloads, History, Local files.
- **Albums from artists you love**, rotating every day.
- Filters All / Playlists / Albums / Artists; list or grid (4×4 pages).
- **Create Aalap playlists**: add songs, rename, delete. Long-press a Spotify playlist to remove it from Aalap only (it stays on Spotify; the account menu brings it back).
- **Artists you follow** with suggestions from what you play.

Player and mini player

- Play/pause, next/previous, shuffle, repeat (all/off), seek, queue (reorder), speed (0.5×–2×), sound/equalizer, devices, share card, lyrics (with karaoke and share lyric), sleep timer, alternative stream, add to playlist, go to album/artist, download.
- **Like**: tap to like (petals fall, button fills, gold *Liked* tag pops), tap again to unlike. **Double-tap the cover** to like with a heart. Likes go to Spotify too.
- Player and queue open as a layer over the page you came from, so pulling down / back / the down arrow slide it into the mini player smoothly.
- Mini player: like with petal rain, swipe to skip, small equalizer on the cover while playing.

Liked Songs

- One list: Spotify + YouTube Music likes (each song once) + songs liked in Aalap. Search, and filters by Spotify / YouTube Music / your top artists. Pull down closes the page.

Profile and Settings

- Liked / Artists / Plays counts; Connected services (Spotify, YouTube Music with *Sync now*, JioSaavn and Gaana as backup sources, Apple Music import).
- **Your taste**: artists you love, languages, plays and how much of each song you finish; updates by itself.
- Appearance: System / Light / Dark. Offline mode, auto-download liked songs, go offline automatically, audio quality, playback options, song videos, log out.

What updates, and how often (verified)

What	When	Seen on the phone
Library sync (Spotify likes and follows, YouTube Music likes, artists, history)	every hour, even when Aalap is closed	job scheduled; YouTube synced "3 minutes ago"
Your taste	after every play (rebuilt at most every 30 s), on every like/unlike, after each hourly sync	2 Diljit Dosanjh plays → his weight went 0 → 0.31; profile rebuilt the same minute
Spotify top artists (taste signal)	daily	was never saved before (bug 1); now saved
Quick picks	daily and on every refresh	date stamp 23 Oct; refresh gave a new set
Aaj ka Aalap 30 songs	daily	23 Oct, 8+7+9+6
Daily Mixes, new releases, Library albums	daily	all stamped 23 Oct
Trending searches	daily, when Search opens	refreshed to 23 Oct on opening Search

What	When	Seen on the phone
Charts	daily	stamped today
Mood stations, Your Mix	every 6 h / 3 h	—
Poster design	midnight	—

3. Test results

PASS = tested on the phone and working (after fixes where noted).

Home

- **PASS** All sections present and in order.
- **PASS** Pull-to-refresh and double-tap Home both give new Quick picks (generation 30 → 31 → 32).
- **PASS** Poster opens today's 30 (8 trending, 7 top, 9 taste, 6 liked); shuffle changes the design and remembers it after restart.
- **PASS** Mood station plays (Romantic → *Nafrat*); equalizer shows on the mini player.
- **PASS** Top 50 India / Global open. **FIXED** Trending India showed the wrong list (bug 2).
- **PASS** Daily Mix plays. **FIXED** mini player flashed "paused" while loading (bug 3).
- **PASS** Artist radio plays (Arijit Singh → *Apna Bana Le*). **FIXED** New releases showed compilations (bug 24).

Search

- **PASS** 10 trending searches refreshed today; typed search works; filters work; no duplicates; tapping plays.
- **FIXED** An artist with no name appeared (bug 4).

Library and playlists

- **PASS** Tiles, filters, grid/list, album rotation, followed artists and suggestions.
- **PASS** Create "QA Test" → add a song → rename to "QA Done" → delete (confirmation shown). Your playlists are back to how they were.
- **PASS** Long-press remove / restore Spotify playlist (tested earlier this session).
- **FIXED** "1 songs" (bugs 5, 7), unhelpful empty Playlists message (bug 6).

Artist and album pages

- **PASS** Artist page (verified badge, listeners, Following, popular songs, releases); album page plays.
- **FIXED** Likes there didn't reach Spotify (bug 8); white tick (bug 9); "Album : " blank (bug 10).

Player

- **PASS** Play, pause, next, previous (screen and system media controls agree), shuffle on/off, repeat on/off, queue, speed, sound/equalizer, sleep timer, add to playlist, alternative stream, share sheet (closed without sending), go to album, go to artist, lyrics + karaoke, close by pull/back/arrow.
- **PASS** Close smoothness on the release build: worst frame 27–32 ms.
- **FIXED** Controls had no screen-reader labels (bug 11); empty "On Spotify" heading (bug 12).

Liked Songs, History, Downloads, Local files

- **PASS** Liked count 195 matches Library and Profile; filters and search work.

- **PASS** History (minutes, top artists/songs), Downloads (1 downloaded song), Local files (import screen).
- **FIXED** History showed "ARIJIT SINGH" (bug 14); Downloads "1 songs" (bug 15); no message for an empty Liked list (bug 17).

Profile, Settings, themes

- **PASS** Connected services, Sync now, Your taste, Appearance.
- **PASS** **Light theme** checked on Home, Search, Library, player (player stays dark on purpose); **Dark** restored.
- **FIXED** Plays 320 vs 327 (bug 13); switches unlabeled and labels not tappable (bug 20).

Background, taste and sync

- **PASS** Hourly sync scheduled; YouTube Music synced recently; play history recording (278 → 330 plays).
- **PASS** Taste updates from use (Diljit Dosanjh test above).
- **FIXED** Spotify top-artists refresh never saved (bug 1); likes didn't refresh taste at once (bug 21).

Offline

- **PASS** Offline mode: Home shows "You're offline" with your downloads, **downloaded song plays**, "Go online" works. Offline mode is off again.
- **FIXED** False "offline" after a network switch (bug 19); auto-offline was off by default (bug 18).

4. Scenarios checked (what happens when things go wrong)

Scenario	What Aalap does now	How checked
No internet / Offline mode	Switches to downloaded music, shows a clear banner, "Go online" button	on phone (Offline mode) + code (auto-offline now on by default)
Network switches (Wi-Fi ↔ mobile)	Re-checks after 2 s and 8 s and on every return to the app, so it doesn't get stuck "offline"	reproduced the stuck state, fixed, code
Spotify unreachable / login expired	Library still shows likes, downloads, history, Aalap playlists and artists, with "Couldn't reach Spotify · Try again"	code (bug 22)
Spotify rate limit (429)	Fails fast; taste uses your library instead; new releases retry later	seen in logs; fixed bug 1
YouTube Music signed out	Liked shows Spotify + Aalap likes; YouTube sync skipped	code
Song can't stream	Skips to the next playable song	code
Phone call during playback	Music pauses (audio focus); unplugging headphones pauses	two real calls during testing + code
App left open past midnight	Poster, Quick picks, today's 30 and mixes roll over	code (bug 16)
New user with nothing liked	Liked page explains how to like songs	code (bug 17)
Empty playlists / albums filter	Friendly message with what to do	on phone (bug 6)

Scenario	What Aalap does now	How checked
Log out	Saved login, live token, cached feeds and welcome step cleared; your plays, Aalap likes, downloads and playlists stay	code (bug 23); not pressed on your phone
Switching Spotify accounts	First sync replaces the old account's likes, follows and taste	code
Large system font	Poster grows instead of overlapping	code
Light / dark theme	Both readable; player always dark	on phone

5. Bugs found and fixed

#	Area	Bug	Fix
1	Taste	Daily Spotify top-artists refresh **never saved anything**: Spotify answers HTTP 429	Falls back to your top artists from Spotify + YouTube Music likes and follows; saved daily
2	Charts	"Trending India" opened Spotify's weekly *Top Songs – India* (Spotify reused the old Viral 50 id); trending tags used it too	Now Spotify's *Trending Now India* (75 songs) for the chart, tags and Aaj ka Aalap
3	Mini player	Showed "paused" for 2–3 s whenever a new song loaded	The internal pause during a track change no longer reaches the button
4	Search	Nameless artist (and possibly songs/albums) in results	Entries without a name are skipped
5	Text	"1 songs" on Library tiles, Liked, Local files, Settings; "1 plays" in Wrapped/History	One helper writes "1 song / 2 songs" everywhere
6	Library	Empty Playlists said "Nothing matches that."	"No playlists yet. Tap + at the top to make one." (and an albums version)
7	Playlists	Header "1 songs · made in Aalap"	Correct count
8	Sync	Likes on **artist pages, album pages and the Liked list** never reached Spotify (and the hourly sync could undo them)	All three mirror to Spotify like the player does
9	Artist page	Liked tick white, saffron everywhere else	Saffron
10	Album page	Header "Album : " with nothing after	"Album · 2024 · 12 songs" (missing parts left out)
11	Accessibility	Player buttons (close, more, shuffle, previous, play/pause, next, repeat, devices, share) had no labels	Labelled for TalkBack
12	Saved in sheet	"On Spotify" heading with nothing under it	Hidden when you have no Spotify playlists to add to
13	Profile	Plays 320 at the top, 327 in Your taste	Both use the engine's count
14	History	Top artist shown as "ARIJIT SINGH"	Prefers the normal spelling
15	Downloads	"1 songs • available offline"	Fixed

#	Area	Bug	Fix
16	Home	Left open past midnight, Quick picks / today's 30 stayed on yesterday	5-minute check rebuilds daily sections when the window changes
17	Liked	New user saw a bare "0 songs"	Explains how to like songs
18	Offline	With no internet the app kept trying to stream (auto-offline off)	Go offline automatically is on by default
19	Offline	App got stuck showing "You're offline" while connected	Debounced detection + re-check on return
20	Accessibility	Settings / Sound / Profile switches unlabeled; tapping the words did nothing	Whole row is the switch, read as "Offline mode, switch, off"
21	Taste	Liking/unliking didn't refresh the taste straight away	Refreshes on every like and unlike
22	Library	If Spotify couldn't be reached, the whole Library vanished	Shows everything on the phone plus a "Try again" note
23	Log out	Two Log out buttons behaved differently; the live Spotify token stayed in memory	One complete log-out used by both
24	Home	"New from your artists" listed compilations/mixtapes ("Happy Vibes", "The Winter Playlist")	Compilations and label mixtapes left out

6. Not run on the phone, and why

Item	Why	Covered instead by
Pressing **Log out** and logging back in	It would sign you out; signing in needs your password, which I don't enter	Code review and fix (bug 23)
Airplane mode / turning Wi-Fi off	Changing phone system settings	In-app Offline mode on the phone + code (bugs 18, 19)
Downloading a song	Writes a file to your phone	Existing download plays offline
Apple Music import, sharing Wrapped	Need your files / would share content	—
Hourly sync firing while the app is closed	Takes up to an hour	Job confirmed scheduled; YouTube sync timestamp recent

7. Known limits and recommendations

1. **Spotify rate limits (429).** Spotify throttles some requests (top artists, top tracks) for web-player tokens. Aalap now degrades gracefully, but a few Spotify-only rows can be missing for a while after heavy use.
2. **Habit score.** The "your habit" part of the recommendation score only counts once you've played the same song-after-song pair before, so it starts at 0 for new pairs and grows with use.
3. **Signing key.** The release build is signed with this computer's debug key (no keystore.properties). Fine for your phone; for sharing or the Play Store, create a release keystore. Updates must be signed with the same key.
4. **App size.** Minify is off (15 MB). Turning on R8 would shrink it, but needs keep rules tested first.

5 New releases cached today still include "Diwali Party Bangers"; the filter for it applies from tomorrow's rebuild.

8. Your phone after testing

Theme **Dark**, Offline mode **off**, shuffle and repeat **off**, music **paused**, test playlist **deleted**, likes unchanged. Two Diljit Dosanjh songs were played for the taste test, so they're in your history. The installed app is **Aalap 3.0.0 (release)**.